

Course Title	Electro-Mechanical Systems	Course Code	EE2010
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	Credit Hrs.	3+1
Knowledge Area	Power Engineering (KA11)	Grading Scheme	Relative
HEC Knowledge Area	Electrical Engineering Core (Breadth)	Applicable From	Spring 2023
Pre-requisite(s)	NL1002, NS1002, EL2004, EE2004		

Course Objective	To analyze the operation of major types of AC and DC machines, including transformers, generators and motors. Quantitative analysis is carried out for efficient engineering application of electrical machines.
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No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
7	An ability to understand the impact of Engineering solutions in the societal and environmental and demonstrate knowledge of and need for sustainable development.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Investigate a transformer circuit using Faraday's Law and electric circuit theory	A1, Q1, M1	C3	1
2	Investigate working of a DC Machine	A2, M1, F	C3	1
3	Analyze Synchronous Generator performance along with special emphasis towards environmental cost of generation	Q2, M2, F	C4	7
4	Appraise the existence of a rotating magnetic field and its application to a Synchronous Motor	Q3, M2, F	C4	1
5	Analyze an Induction motor circuit	A3, F	C4	1

Text Books	Title	Electrical Machinery fundamentals
	Author	Stephen J. Chapman
	Publisher	McGraw Hill, 2005, 4th Edition, ISBN 0-07-246523-9
Reference Books	Title	Electric Machinery
	Author	Fitzgerald, A.E., Charles Kingsley, Jr. And Stephen Umans
	Publisher	Boston: McGraw Hill, 2002, 6th Edition

Week	Course Contents/Topics	Chapter*	CLO*
01	Fundamentals of Electricity and Magnetism: B-H curve, Hysteresis	1	1
02	Ideal Transformers: Applied and induced voltage, Circuit symbol, problems	2	1
03	Practical Transformers: Equivalent circuit, Polarity test, Taps, Losses, problems	2	1
04	Practical Transformers: Transformer tests, efficiency and voltage regulations, problems	2	1
05	Three phase transformers: Basic properties of 3-phase transformers, $\Delta\Delta$, ΔY , $Y\Delta$ and YY connections, autotransformers	2	1
06	Direct Current Generators: Theory and Equivalent circuits	9	2
07	Direct Current Motors: Principle, Back EMF, construction, equivalent circuits.	09	2
8	Direct Current Motors: Types, Speed control, Torque Speed Curve, Problems	09	2
9	Synchronous Generators: Introduction, Commercial synchronous generators, Number of poles, stator, rotor, Field excitors, Brushless excitation,	04	3
10	Synchronous Generators: Armature Reaction, Equivalent circuit, Efficiency and Voltage Regulation	05	3
11	Synchronous Motors: Introduction, Construction, Starting, Pull-in torque, etc	06	4
12	Synchronous Motors: Power and torque, Excitation and reactive power,	06	4
13	Three-phase Induction Motors: Introduction, components, Principle of operation, The rotating field, Direction of rotation,	07	5
14	Three-phase Induction Motors: Starting, Torque Speed Curve, Creating High starting torque	07	5
15	Single-phase Induction Motors: Introduction, Construction, Synchronous speed, Torque-speed characteristic, Principle of operation, Capacitor-start motor	11	5
16	Problem solving	07,11	5

*Reference book chapters are given in brackets

Assessment Tools	Weightage
Quizzes, Assignments	20.0%
Midterm (I+II)	30.0%
Final Exam	50.0%